

Phase 6: Project Testing

OptiCrop: Smart Agricultural Production Optimization

Role	Name
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Model Evaluation

During the training phase, multiple models were evaluated. Logistic Regression was selected for deployment due to its balance of high accuracy and lightweight inference. The final metrics on the test set were:

- **Accuracy:** ~96%
- **Precision (Weighted Avg):** ~0.96
- **Recall (Weighted Avg):** ~0.96
- **F1-Score (Weighted Avg):** ~0.96

Random Forest achieved ~99% but was deemed unnecessarily heavy for this specific web deployment.

System Verification Cases

Black-box testing was performed on the Flask application to ensure robustness against various user inputs.

Input Scenario	Expected Outcome	Actual Output	Status
Valid Rice Parameters (N:90, P:42, K:43, Temp:20.8, Hum:82, pH:6.5, Rain:202)	Crop: Rice	Crop: Rice	PASS
Valid Maize Parameters (N:80, P:40, K:40, Temp:25, Hum:60, pH:7, Rain:100)	Crop: Maize	Crop: Maize	PASS
Empty Form Submission	Error message to fill all fields	Error message displayed	PASS
Invalid Data (Text instead of numbers)	HTTP 400/Error handling trigger	Error caught by try-except block	PASS